

Parent Guide — senior / module-05-dot-addition

IKS Curriculum

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Parent Guide — Module 5: Bindu Paddhati (Senior)

Your student's age: Senior (Grades 9–12) · **Module length:** 10 sessions × 45 min

What your student is learning

Two equally important things:

1. **The dot method at high speed** — sub-60-second mental addition on 5-addend × 5-digit columns, plus extending the technique to base 8, base 16, and even binary.
2. **The historicity question** — was “Vedic mathematics” actually Vedic? They read Bharati Krishna Tirthaji's preface (1965) and S. G. Dani's scholarly critique (1993), then form their own informed position.

The honest framing

Your student will learn — and articulate — these uncomfortable facts: - The dot method is **not** found in any Vedic text. - Tirthaji claimed his 16 sūtras came from an *Atharva-veda Pariśiṣṭa* (Vedic appendix-text) that no other scholar has ever produced. - S. G. Dani (TIFR / IIT Bombay) argues in *Frontline* (1993) and *Resonance* (2001) that the techniques are largely 20th-century innovations parallel to similar Western mental-arithmetic systems (Trachtenberg). - The math is **real and useful**. The Vedic-provenance claim is **contested**.

This is not anti-Indian-culture. It is honest scholarship about a specific 20th-century intellectual project. Modern postural yoga has a similar history (covered in Module 8 Senior).

Project: choose your track

5–6 page paper. Two options:

Math track: Generalise the dot method to base- n and prove correctness via place-value reasoning. Optionally extend to multiplication (*Ūrdhva-Tiryagbhyām*).

History track: Argue a position on the labelling debate — keep “Vedic mathematics,” drop it, or retain with caveat. Or place Tirthaji's project in its post-colonial historical context.

How to help

Helpful: - Be a willing test subject for their data project if they choose that. - Read their draft and ask: “Which point would a skeptical reader push back on?” - For the history track: ask them to articulate Dani’s strongest critique fairly. If they strawman Dani, push back.

Not helpful: - Insisting “Tirthaji was right” or “Dani was right.” The point is *informed engagement*, not picking a winner. - Confusing this module with anti-Indian-tradition. The dot method is a real Indian intellectual achievement; we just don’t pretend it’s older than it is.

On the politics

Some families teach “Vedic mathematics” with pride and may feel this honest framing diminishes that pride. The classroom answer: *cultural achievements are not less valuable for being recent*. Modern postural yoga, modern Vedic mathematics, modern Indian classical music — all real, all Indian, just younger than their names suggest.

Assessment

- Summative quiz: 35 marks.
- Paper: 65 marks (math track and history track have different rubric weightings).
- Total: 100 marks.

What success looks like

By end of module, your student can: - Add 5×5-digit sums in under 60 seconds using the dot method. - Apply the dot method in non-decimal bases (8, 16, 2). - State and explain Tirthaji’s 16 sūtras. - Articulate Dani’s three main critiques of the Vedic-provenance claim. - Defend an informed position on the labelling debate, with citations and a steel-manned counter-argument.

Want to go further?

After this module, suggested reading: - Tirthaji, B. K. (1965). *Vedic Mathematics: Sixteen Mathematical Formulae from the Vedas*. - Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” *Frontline*. - Plofker, K. (2009). *Mathematics in India*. (Standard history.) - Hayashi, T. (1995). *The Bakhshali Manuscript*. (For pre-Tirthaji Indian arithmetic.)

10-Day Lesson Plan

Module 5 (Senior) — 10-Day Lesson Plan

At a glance

Day	Title	File	Output
1	Baseline + Historicity	days/day-01-hook.md	Self-baseline + opening question

Day	Title	File	Output
2	The 16 Sūtras	days/day-02-sutras.md	Sūtra-recall quiz
3	Dot Method (Senior)	days/day-03-dot-method.md	Speed check
4	Sūtra Applications	days/day-04-sutra-apps.md	Formative
5	Base-n Generalisation	days/day-05-base-n.md	Formative quiz Part B
6	Historicity	days/day-06-historicity.md	Reading response
7	Discussion	days/day-07-discussion.md	Oral assessment
8	Project Session 1	days/day-08-project-session-1.md	Outline
9	Project Session 2	days/day-09-project-session-2.md	Draft
10	Final	days/day-10-final-assessment.md	Summative + paper

Pacing notes

- Days 1–5 are math-and-procedure heavy. Days 6–10 are reading and writing heavy.
- The Dani 1993 reading on Day 6 is the most challenging; budget 20+ minutes for it.
- Project tracks (math vs history) split the class — plan supports for both during Days 8–9.

What students leave with

- Sub-60-second 4×4 column addition by dot method.
- Working knowledge of all 16 Tirthaji sūtras.
- Ability to extend the dot method to non-decimal bases.
- An informed position on the historicity of “Vedic mathematics” claims.
- One research paper.

Homework

Homework — Module 5 (Senior)

Night	Task	Time
Day 1 → 2	Read Tirthaji’s preface. Note 1 question.	20 min

Night	Task	Time
Day 2 → 3	Memorise the 16 sūtras (Sanskrit names only).	20 min
Day 3 → 4	Practise 10 dot-method problems at home.	15 min
Day 4 → 5	Review sūtras 7 & 8 worked examples.	15 min
Day 5 → 6	Prep for Dani reading — review provenance question.	15 min
Day 6 → 7	Choose your debate position; prep 2-min argument.	30 min
Day 7 → 8	Submit project plan: track + thesis + 3 sources.	30 min
Day 8 → 9	Draft 1.5–2 pages.	60–90 min
Day 9 → 10	Complete draft; proofread; format.	60–90 min